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MILITARY SPECIFICATION

GRINDING MACHINES, TOOL AND CUTTER; AND GRINDING AND GASHING MACHINES, TOOL, CUTTER, FLUTE AND RADIUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, electric motor driven, manual and power operated tool and cutter grinders and tool, cutter, flute and radius grinders with equipment and accessories specified herein.

1.2 Classification. The grinding machines shall be of the following types, styles and sizes. The type, style and size to be furnished shall be as specified (see 6.2.1).

Type I - Floor mounted, horizontal spindle, tilting and swiveling wheelhead tool and cutter grinder.

Style A - Manual.

Size	Swing over table (diameter) (inches)	Maximum distance between tailstock centers (inches)
1	8	15, 22*, 24*, 32*
2	10	17*, 22, 25*, 27*, 33*
3	12	27, 32*

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Style B - Power operated.

Size	Swing over table (diameter) (inches)	Maximum distance between tailstock centers (inches)
1	8	15, 17*, 22*, 24*, 32*
2	10	22, 25*, 27*, 33*
3	12	21*, 27, 32*

Type II - Floor mounted, vertical spindle tool and cutter grinder.

Style A - Manual.
Style B - Power operated.

Size	Swing over table (diameter) (inches)	Maximum distance between tailstock centers (inches)
1	8	15

Type III - Floor mounted, horizontal spindle, reciprocating wheelhead, manual tool, cutter, flute and radius grinder.

Size	Maximum workpiece diameter between centers (inches)	Maximum workpiece length between centers (inches)
1	12 3/4	7

Type IV - Floor mounted, horizontal spindle, traversing wheelhead, manual tool, cutter, gashing, flute and radius grinder.

Size	Swing over main slide (diameter) (inches)	Maximum workpiece length between centers (inches)
1	12	6

* Indicates optional sizes available.

When optional or other sizes are required, the machine size and its corresponding characteristics shall be as specified (see 6.2.1).

2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001).

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NEMA ICS 1 - Industrial Control and Systems.

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ANSI/NEMA ICS 6 - Enclosures for Industrial Controls and Systems.

ANSI/NFPA 70 - National Electrical Code.

ANSI/NFPA 79 - Electrical Standards for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B5.10 - Machine Tapers.

ANSI B7.1 - Abrasive Wheels, Use, Care, and Protection of, Safety Requirements for the.

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines.

ANSI B74.2 - Shapes and Size of Grinding Wheels, and Shapes, Sizes, and Identification of Mounted Wheels, Specifications for.

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

ANSI B94.19 - Milling Cutters and End Mills.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3351).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187).

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

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(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3351).

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models. The machine shall be capable of resharpening conventional type cutters and producing new cutters, when required, to tolerances and finishes in accordance with the requirements herein. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Machine types and styles.

3.2.1.1 Type I. Type I machines shall be floor mounted tool and cutter grinders. Style A machines shall be manually operated with handwheels or levers provided to allow manual control of table transverse and longitudinal travel. Style B machines shall have power table travel with handwheels or levers provided for manual set-up and adjustment. The wheelhead spindle shall be mounted horizontally. The workhead shall be either a manual universal workhead or a motor driven independent workhead. Type I machines shall be equipped with attachments for external and internal cylindrical grinding, gear cutter grinding, and surface grinding as specified herein. Type I machines shall sharpen a variety of cutters including milling cutters, face mills, end mills, reamers, gear cutters, counterbores and other metal cutting tools.

3.2.1.2 Type II. Type II machines shall be floor mounted tool and cutter grinders. The machines shall be either manually operated with handwheels and levers or power operated using pushbutton controls. The wheelhead spindle shall be vertically mounted. The workhead shall be a manual universal

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fixture. Type II machines shall be equipped with attachments for radius grinding and broach grinding as specified herein. Type II machines shall sharpen a variety of cutters including milling cutters, face mills, end mills, reamers, gear cutters and other metal-cutting tools.

3.2.1.3 Type III. Type III machines shall be floor mounted tool, cutter, flute, and radius grinders. The machines shall be manually operated with handwheels or levers. The wheelhead spindle shall be horizontally mounted. The workhead shall be either a manual universal workhead or a motor driven independent workhead. Type III machines shall sharpen cutting tools having radii in multiple cutting surfaces and grind all clearances on profile, straight, or angular cutters. Type III machines shall sharpen a variety of cutters including milling cutters, shell end mills, ball end mills, and other metal-cutting tools with convex or concave radii and straight and spiral flutes.

3.2.1.4 Type IV. Type IV machines shall be floor mounted tool, cutter, gashing, flute, and radius grinders. The machine shall be manually operated with handwheels or levers. The wheelhead shall be horizontally mounted. The workhead shall be a motor driven independent workhead with a spiral and relief grinding attachment. Type IV machines shall resharpen conventional types of cutters including end mills, reamers, counterbores, and form cutters with straight, tapered, or helical teeth and involving convex or concave radii. Type IV machines shall also make new cutters from solid stock.

3.2.2 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2.2 Digital readout system. When specified (see 6.2.1), a digital readout system shall be provided on the X and Y axes of travel to indicate longitudinal worktable and cross slide position. The digital readout system

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shall consist of a position indication device, an electronic display for each axis and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall be capable of providing a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The system accuracy shall be 0.0002 inch per foot and the repeatability shall be 0.0001 inch. Special requirements for digital readout system components shall be as specified (see 6.2.1). The readout system shall be capable of indicating axis location in both the US and SI systems of measurement.

3.2.2.2.1 Position indication device. A position indication device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The indication device shall be of the independent type. Each indication device shall be installed with a cover or guard to prevent physical damage to any portion of the device. The device shall provide axis position indication over the entire range of axis travel for each specified axis. Provisions shall be made to prevent damage to the indication device in the event of overtravel.

3.2.3 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.4 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the latest developments available within the industry.

3.2.5 Operating controls. The machine shall be completely wired through the controls and ready for connection to the applicable power source. Unless otherwise specified (see 6.2.1), the controls shall conform to ANSI/NEMA ICS 1 and shall be installed in enclosures conforming to ANSI/NEMA ICS 6. Unless otherwise specified (see 6.2.1), the machine's on-off switch shall incorporate overload protection if not provided in the motor or control. Controls for all functions of the machine shall be arranged in a location convenient to the operator. Controls, such as push buttons, dials, and switches, shall be functionally arranged and clearly identified to facilitate use.

3.2.6 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the grinding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically

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required herein, shall comply with all of the requirements of ANSI B11.9 and OSHA 29 CFR 1910 that are applicable to the machine itself. In the event of a conflict between the requirements of ANSI B11.9 and OSHA 29 CFR 1910, the requirements of OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.7 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.9 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.10 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.11 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to indicate stock removal, to position machine axes, or for tool movement shall be graduated in increments not larger than 0.001 inch or less. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than 45 Rockwell C.

3.3.9 Machine ways. All machine ways shall be adequately protected regardless of the position of machine components to exclude dust, dirt, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members. The worktable ways shall be hardened to 40 Rockwell C and ground to 30 microinch. All other machine component ways shall either be also hardened to 40 Rockwell C and ground to 30 microinch or hand scraped.

3.3.10 Slides. All sliding members shall be provided with means of adjustment to meet and maintain the accuracy and alignment tolerances given in table III or IV, as applicable for the machine type, style, and size specified.

3.3.11 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. An adjustable wheel guard(s) shall be furnished for the grinding wheel(s).

3.3.12 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

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3.4 Components. The machine shall consist basically of the following components, as applicable to the type and style machine specified in 1.2. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. Each machine base shall be a one- or two-piece cabinet type casting or stress-relieved weldment of iron or steel, ribbed and braced to minimize deflection and vibration under maximum operating loads and to ensure accuracies required herein. When applicable, coolant reservoirs or dust extractors maybe housed in the base or provided as separate external units. An access door(s) or removable plate(s) shall be provided as necessary to service components housed within the base. Means shall be provided for floor mounting and leveling of the machine.

3.4.1.1 Type I base. The base on type I machines shall be constructed with proportional V-ways, flat ways or equal means to accept either a stationary wheelhead column and a saddle to support a traversing, sliding, swiveling worktable arrangement or a saddle to support a traversing wheelhead column and a sliding, swiveling worktable arrangement.

3.4.1.2 Type II base. Type II machines shall be constructed with a pedestal integrally attached to the base to support the vertical grinding wheelhead assembly. The base shall be constructed to also mount a vertically adjustable compound worktable column.

3.4.1.3 Type III base. The base on type III machines shall be constructed with proportioned flat or steel V-ways to accept the main wheelhead slide. The base shall be constructed to also mount a rotating compound worktable arrangement.

3.4.1.4 Type IV base. Type IV machine bases shall be constructed with proportioned steel flat or V-ways to accept a traversing main wheelhead slide. The base shall be constructed to also mount a rotating compound worktable arrangement.

3.4.2 Worktable arrangement. Unless otherwise specified (see 6.2.1), all worktable component travel and characteristics shall meet the requirements given in table I or table II, as applicable, for the machine type, style, and size.

3.4.2.1 Type I worktable arrangement. The saddle on type I machines shall be mounted on base ways and shall travel transversely. The saddle shall support either the wheelhead column or the sliding, swiveling worktable arrangement. A sliding table shall be mounted either directly on the base or on the saddle and shall travel longitudinally. A T-slot and adjustable dogs or a table dog rack with adjustment screws shall be provided to reverse table motion. A minimum of two dogs shall be provided. The top of the sliding table shall be machine finished to accept a swivel table. The swiveling workholding table shall be provided with graduations to indicate

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table swivel in degrees and inches per foot. A fine screw adjustment shall be provided for taper settings up to not less than 2 inches per foot, indicated on a scale at the end of the table. Means shall be provided to lock all moving components after positioning. The top of the swiveling workholding table shall be constructed with a T-slot conforming to ANSI B5.1 or comparable means for attaching a workholding device. The saddle transverse feed mechanism shall be hand-operated with a fine screw adjustment having a dial conforming to 3.3.7. Table longitudinal feed on style A machines shall be manually operated utilizing handwheels or levers. Table longitudinal feed on style B machines shall be either manually operated feed, utilizing handwheels, or infinitely variable, power operated feed, utilizing pushbuttons. Controls shall be provided for selecting rates of power longitudinal travel on the table. Means shall be provided for selecting fine and coarse crossfeed on all machines.

3.4.2.2 Type II worktable arrangement. Type II machines shall be designed with a vertically adjustable worktable column supported by the base. A handwheel shall be provided for manually raising and lowering the column. Covers or bellows shall be provided on the adjustable worktable column to protect it from contaminants in any position. A swiveling, compound worktable shall be mounted on the adjustable column. The compound worktable shall swivel about a fixed center and be provided with graduations for indicating table swivel in degrees. The compound worktable shall also provide longitudinal and transverse travel for machine set-up. The table top shall be provided with three T-slots conforming to ANSI B5.1 or comparable means for mounting an angular work support fixture that can be used for supporting the universal work spindle or single T-slotted vertically swiveling table. Means for clamping the adjustable column and swiveling table after positioning shall be provided. Transverse travel shall be provided by a wheelspindle column supported on a ram on a column which moves the wheelspindle into the work. Each machine shall have ample bearing surface to assure permanently accurate alignment. Longitudinal and rotational worktable and saddle motion shall be manually operated by handwheels having graduated dials conforming to 3.3.7. When specified (see 6.2.1), automatically controlled machine cycle with handwheels available for manual machine set-up shall be provided. The automatic cycle shall allow the machine after set-up to complete roughing passes, finishing or spark out passes and sheet the machine off when the cycle is completed.

3.4.2.3. Type III worktable arrangement. The work holding unit on type III machines shall be supported by a turntable mounting which shall allow swiveling the work holding unit in the horizontal plane. Adjustable stops shall be provided to limit the swivel in both directions. The turntable mounting shall mate directly with the base. The swiveling turntable shall carry a screw-operated cross-slide to allow offsetting the axis of the workhead spindle from the center of the turntable. On the cross-slide shall be mounted a second slide, known as the top slide, which shall carry the work holding unit. The top slide shall also be screw-operated and shall allow adjustment of the work holding unit in front and behind the center of the swiveling table. Stop dogs and dials graduated in accordance with 3.3.7 shall be provided to allow accurate positioning of the work holding

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unit slides. The top slide shall permit the work holding unit to be adjusted to suit the length of work. Type III machines shall have a work holding unit mounted on the top slide. Means shall be provided to lock all moving components after positioning. All slides shall be provided with a one-shot lubrication system.

3.4.2.4 Type IV worktable arrangement. The work holding unit on type IV machines shall be supported by a turntable mounting which shall allow swiveling the work holding unit in the horizontal plane. Adjustable stops shall be provided to limit the swivel in both directions. The turntable mounting shall mate directly with the base. The swiveling turntable shall carry a screw-operated cross-slide to allow offsetting the axis of the work-head spindle from the center of the turntable. On the cross-slide shall be mounted a second slide, known as the top slide, which shall carry the work holding unit. The top slide shall also be screw-operated and shall allow adjustment of the work holding unit in front and behind the center of the swiveling table. Micrometer stop dogs and dials graduated in accordance with 3.3.7 shall be provided to allow accurate positioning of the work holding unit slides. The top slide shall permit the work holding unit to be adjusted to suit the length of work. A positive stop shall be provided to bear against the rear surface when the head is centrally located. Type IV machines shall have a work holding unit housing mounted on the top slide which shall allow the unit to be tilted above and below a horizontal position. Means shall be provided to lock all moving components after positioning. All slides shall be provided with a one-shot lubrication system. All slide movement shall be manually controlled using handwheels or levers.

3.4.3 Wheelhead. Unless otherwise specified (see 6.2.1), the wheelhead unit shall meet the travel and characteristic requirements given in table I or table II, as applicable, for the machine type, style, and size.

3.4.3.1 Type I wheelhead. The grinding wheelhead on type I machines shall be of the self-contained type. The spindle shall be of high grade steel, hardened and ground, and mounted on precision, preloaded, antifriction bearings with sealed-in lifetime lubrication or cartridge-type mounting. The spindle shall be designed for using straight, cup, or dish wheels. The wheelhead shall swivel in a horizontal plane at least 106° both sides of zero. The wheelhead shall tilt at least 15° above and below center. Wheelhead swivel and tilt shall be manually set with dials or scales graduated in 1° increments or less to indicate the angular setting. The wheelhead shall give vibrationless operation up to the maximum rated RPM of the grinder. Unless otherwise specified (see 6.2.1), two or more changes of wheel speed shall provide 3,800 to 5,500 RPM using 3-1/2 to 6 inch diameter wheels. The spindle shall be designed to permit removal of the spindle assembly from the machine as a unit for servicing. The wheelhead shall be mounted on either a wheelhead column mounted on the base or a wheelhead saddle slide and shall be vertically adjustable using an elevating handwheel graduated in accordance with 3.3.7. A wheel spindle motor shall be mounted in or on the wheelhead and shall be provided with means for maintaining belt tension, if a timing belt is not used. The belt shall be guarded by a suitable cover.

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3.4.3.2 Type II wheelhead. The wheelhead unit on type II machines shall be of the self-contained type. The spindle shall be of high grade steel, hardened and ground, and mounted on precision, preloaded antifriction bearings with sealed-in lifetime lubrication or cartridge-type mounting. The spindle shall be designed for using cup wheels. Two or more changes of wheel spindle speed shall provide wheel speeds from 3,200 to 4,800 RPM using 4, 5, or 6 inch diameter wheels. The wheelhead shall be mounted on a reciprocating ram and the grinding wheel spindle shall be mounted vertically. When specified (see 6.2.1), an optional horizontal grinding wheel spindle shall be provided. The grinding wheel shall be vertically adjustable for set-up. The wheelhead shall also be horizontally adjustable on the ram and shall tilt 90° left of zero and 70° right of zero. Dials or scales graduated in 1° increments or less shall be provided to indicated angular settings. A wheel spindle motor shall be mounted in or on the wheelhead and shall be provided with means for maintaining belt tension, if a timing belt is not used. A protective cover shall be provided for the belt. When automatic cycle is provided (see 3.4.2.2), the cycle shall allow the operator to establish such parameters as ram reciprocation, cutter index, wheel spindle rotation, and feed and work spindle traverse and place the machine in automatic mode to operate unattended during a machine cycle.

3.4.3.3 Type III wheelhead. On type III machines, the wheelhead unit shall consist of sliding members, permitting transverse and longitudinal adjustment. A main slide shall mate with the base and provide longitudinal adjustment. A wheelhead cross slide shall be mounted on the main slide and shall provide transverse adjustment of the wheelhead. The wheelhead unit itself shall be mounted on the cross slide in T-slots. All wheelhead adjustment shall be manually controlled. The slides shall be provided with lubrication. Scales shall be provided to indicate longitudinal and transverse adjustment. The grinding wheel shall be mounted on an individual spindle supported on precision, preloaded, antifriction bearings with sealed-in lifetime lubrication or cartridge-type mounting. The spindle shall be designed for using straight wheels. Two wheel spindle speeds shall provide 3,800 to 5,800 RPM wheel speed using 4 to 6 inch diameter wheels. The spindle shall provide vibrationless operation up to the maximum RPM of the grinder. The grinding wheel motor shall be mounted in or on the wheelhead and shall be provided with means for maintaining belt tension, if a timing type belt is not used. A protective cover shall be provided for the belt.

3.4.3.4 Type IV wheelhead. On type IV machines, the wheelhead unit shall consist of three sliding members providing longitudinal, transverse, and vertical adjustment of the unit. A wheelhead slide shall mate with the base and provide transverse movement. Mounted to this slide shall be another slide and vertical column to provide longitudinal movement and vertical adjustment. The slides shall be provided with means for lubrication. Longitudinal, transverse, and vertical adjustment shall be manually operated by handwheels or handcranks, graduated according to 3.3.7. The wheelhead unit shall be fitted with a permanently mounted clearance angle scale to

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provide quick setting of the grinding wheel for any desired clearance angle in both the vertical and transverse directions. The grinding wheel shall be mounted on an individual spindle supported on precision, antifriction bearings. The spindle shall be designed for using straight, dish, and cup wheels. The spindle shall provide vibrationless operation of the grinder up to its maximum rated RPM. Three spindle speeds shall provide 4,000 to 8,000 RPM wheel speed using 3-1/2 and 4 inch diameter wheels. The grinding wheel motor shall be mounted on the vertical slide and be belt connected to the wheel spindle. The belt shall be guarded by a suitable cover.

3.4.4 Workhead.

3.4.4.1 Type I workhead. Unless otherwise specified (see 6.2.1), type I machines shall be provided with a manually operated, universal workhead capable of swiveling 360° in a horizontal plane and tilting at least 90° in a vertical plane. The workhead shall be mounted on the swiveling workholding table (see 3.4.2.1). All angular settings shall be indicated on dials or scales graduated in increments of 1° or less. The workhead spindle shall be of high-grade steel, hardened and ground, and shall run in antifriction bearings. Spindle tapers shall be manufacturer's standard. The workhead shall be designed so it can be accurately aligned on the table and securely held in position. The workhead shall be complete with center and draw-in bolt.

3.4.4.2 Type II workhead. Unless otherwise specified (see 6.2.1), type II machines shall be provided with centers and arbor mounted on a subtable and a manually operated, universal workhead. Both the centers, subtable and universal workhead shall mount in an angular work support fixture capable of tilting 180° in a vertical plane. The angular work support fixture shall mount in the worktable T-slots (see 3.4.2.2). All angular settings shall be indicated on dials or scales graduated in increments of 1° or less. The workhead spindle shall be high-grade steel, hardened and ground, and shall run in antifriction bearings. Tapers shall be manufacturer's standard. The workhead shall be complete with center and draw-in bolt.

3.4.4.3 Type III workhead. The workhead on type III machines shall be mounted on a vertical master slide bolted on the worktable slide of the worktable arrangement (see 3.4.2.3). The workhead shall be vertically and transversely adjustable on the vertical and worktable slides. Unless otherwise specified (see 6.2.1), the workhead shall be a manually operated, universal workhead capable of pivoting the axis of the cutter $\pm 40°$ from horizontal. The workhead shall revolve about the axis of the cutter an amount equal to the helix angle of the cutter. Machine design shall permit making a gash to center with negative, positive, and radial rake with adjustments to secure the desire rake. The work holder shall have means for indexing 2, 3, 4, 6, and 8 flute mills. A tooth rest shall be provided for mills with more than 4 flutes. Other plates shall be required for mills other than described. Graduations shall be provided for setting clearances desired when grinding end mills. A work holding spindle shall be mounted on

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antifriction bearings and shall accept adapters for reducing collets, as specified in 3.8.4(f). Tapers shall be manufacturer's standard and shall conform to ANSI B5.10.

3.4.4.4 Type IV workhead. The workhead unit on type IV machines shall be mounted on the main slide of the worktable arrangement with a suitable mechanism operating through a spiral lead device. A set of index plates for 8, 10, and 12 equally notched spaces shall be provided. A workhead spindle shall also be fitted with a permanently mounted, direct indexing disk, a spring-loaded micrometer, and an adjustable index finger for ratchet indexing of 10, 12, and 14 evenly spaced notches. The spindle shall be mounted on precision, antifriction bearings and shall be made to accommodate split collets. An adjustable tooth rest shall be provided for mounting on the grinding wheel spindle housing. The workhead unit shall be equipped with a permanently mounted dust protected motor for effecting rotation of the spindle by belt drive. A set of three index plates for a helical lead mechanism to index 8, 10, and 12 divisions shall be furnished with the workhead unit.

3.4.5 Headstock. Type I, style B machines shall be provided with a motor driven, independent headstock unit. The headstock shall be independent of the wheelhead. It shall be adjustable along the table and removable from the table. It shall have a horizontal tilt adjustment of not less than 90° both sides of zero alignment with the table, indicated by graduations in degrees. The headstock spindle shall be of alloy steel, shall be ground and have a surface hardness of not less than Rockwell 48 on the C scale. It shall be mounted in preloaded ball or roller bearings, arranged for either dead or live center grinding. The taper shall be the manufacturer's standard, fitted with a center. The spindle shall be arranged for mounting faceplates and chucks. The headstock shall meet the specifications given in table I. The headstock shall be furnished with a live center pulley.

3.4.6 Tailstock. When specified (see 6.2.1), a tailstock(s) shall be provided with the machine. The tailstock shall travel along the table in alignment with the headstock, when a headstock is provided and set on zero. The tailstock taper, or other internal configuration, shall be as specified (see 6.2.1). The spindle shall act against a spring to allow tension adjustment. A clamping device shall be provided for locking the spindle with desired tension.

3.4.7 Coolant system. When specified (see 6.2.1), a flood type or mist type coolant system shall be provided. The system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit full flow of coolant, and shall have means for draining and cleaning. The system shall include a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant flow.

3.4.8 Hydraulic system. When specified (see 6.2.1), a hydraulic system shall be provided. The system shall be complete including all pumps,

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valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a clean-out access plate if the reservoir is not removable.

3.4.9 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.9.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements of ANSI/NEMA MG 1 and ANSI/NFPA 70.

3.4.9.2 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.4.10 Dust collector. When specified (see 6.2.1), a dust collection unit shall be provided. The dust collector shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel. The dust collector shall be either a built-in unit or a separate external unit.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in tables I, II or III for the machine type, style, and size specified.

3.6 Performance. Each machine shall be capable of resharpening conventional type cutters to like new condition tolerances and finishes. The machine shall grind workpieces in accordance with the test procedures given in 4.7.1 and 4.7.2. Finished cutter physical dimensions shall conform to ANSI B94.19 for the type of tool specified to be ground. Diameters of cutter tooth edges, when measured at any point along their full length, shall be within +0.000 inch to -0.0015 inch. Runout shall not exceed 0.0015 inch TIR. Surface finish shall not be rougher than 16 microinch.

3.6.1 Additional performance requirements. Performance requirements required in addition to, or in lieu of, those given herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances given in tables IV and V when tested using the procedures given in 4.7.3.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following standard equipment shall be furnished for the machine type, style, and size specified.

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3.8.1 Type I standard equipment.

a. Three grinding centers.

b. Set of work driving dogs.

c. Left- and right-hand tailstocks with centers.

d. All tooth rests, tooth rest bars, holders, tooth rest holder clamps, blades, and other tooth rest equipment normally furnished with the grinder.

e. Wheel sleeves, mounts, collets, or mandrels normally furnished with machine.

f. Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

g. Center height gage.

h. Reducing collets, the number of each and sizes shall be as specified (see 6.2.1). Tapers shall match universal workhead and headstock (see 3.4.4.1 and 3.4.5).

i. All wrenches, cranks, handles, clamps, cutter bars, wheel guards, wheel pullers, T-bolts, nuts, and washers normally furnished with the grinder.

j. One diamond of at least 0.20 carat, complete with holder and bracket.

3.8.2 Type II standard equipment.

a. Attachment equipped with centers, fitted to the universal work support fixture.

b. Universal tooth rest, plain tooth rest, blades, and all other tooth rests furnished commercially with machine.

c. Centering gage.

d. Wheel mounts, mandrels, collets, or sleeves, normally furnished with machine.

e. All wrenches, cranks, handles, wheel guards, T-bolts, nuts, and washers, normally furnished with the machine.

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f. Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

g. One diamond of at least 0.20 carat, complete with holder and bracket.

h. Reducing collets, the number of each and sizes shall be as specified (see 6.2.1). Tapers shall match universal work support fixture.

3.8.3 Type III standard equipment.

a. All necessary wrenches, cranks, handles, drive pulleys, drive belts, wheel guards, T-bolts, nuts, and stock supports normally furnished as standard equipment with the machine.

b. Cutter tooth setting gage.

c. Universal tooth rest and holder.

d. Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

e. One diamond of at least 0.20 carat, complete with holder and bracket.

f. Ball bearing work holding spindle bored one of the following ways, as specified (see 6.2.1):

(1) For #11 B&S taper shank, with 11 to 9 and 11 to 7 reducing collets.

(2) As required by the procuring agency.

g. Work light, complete with protective shield and on-off switch.

3.8.4 Type IV standard equipment.

a. All necessary wrenches, cranks, handles, drive pulleys, drive belts, wheel guards, T-bolts, nuts, and stock supports normally furnished as standard equipment with the machine.

b. Tooth rest assembly with one extension bar.

c. Tooth rest centering gages.

d. Set of three index plates for helix lead mechanism indexing device, 8, 10, and 12 divisions.

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e. Set of 13 straight cylindrical collets with adapters as required 1/8 inch, 3/16 inch, 1/4 inch, 5/16 inch, 3/8 inch, 7/16 inch, 1/2 inch, 5/8 inch, 3/4 inch, 7/8 inch, 1 inch, 1-1/8 inches, and 1-1/4 inches.

f. Set of six taper collets - the O.D of collets shall fit the workhead spindle. The size of work holding taper shall be as specified (see 6.2.1).

g. Grinding wheels, number of each wheel size, type, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

h. One diamond of at least 0.20 carat, complete with holder and bracket.

i. Centering gage.

3.9 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1). Optional equipment shall be adaptable and suitable for the type, style, and size machine furnished.

3.9.1 Type I optional equipment.

a. One cylindrical grinding attachment, complete with 3-, 4- or 6-jaw universal chuck, two sets nonreversible chuck jaws (one internal and external) chuck adapter, spindle assembly, driving motor and all other associated components required. Chuck size, spindle taper, and motor horsepower shall be as specified (see 6.2.1).

b. One internal grinding attachment complete with mounting bracket, wheel spindle housing and spindle (for use with removable quills) driven by wheelhead spindle through a pulley and belt or self-contained motor. The number of quills and their sizes, and the number of grinding wheels, their sizes, types, dimensional characteristics, and material composition shall be as specified (see 6.2.1).

c. One surface grinding attachment, complete with swivel vise (360° in both horizontal and vertical planes), intermediate support, base plate, and necessary T-bolts.

d. One gear cutter sharpening attachment including bracket, gage, pawl, necessary T-bolts, and cutter bushings. The number and size cutter bushings shall be as specified (see 6.2.1).

e. Radius grinding attachment with radius and maximum cutter capacity as specified (see 6.2.1), including swivel housing and table, top plate, workhead support and locating gage.

f. Other equipment as specified by the procuring activity.

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3.9.2 Type II optional equipment.

a. One radius grinding fixture for cutters with a maximum diameter of 6 inches and radii up to 1-1/2 inches with a #12 B&S taper or #50 international taper as specified (see 6.2.1).

b. One broach grinding fixture for flat broaches up to 3-3/4 inches wide.

c. One tool vise workholding fixture.

d. Other equipment as specified by the procuring activity.

3.9.3 Type III optional equipment.

a. One center assembly to mount tools between centers.

b. One motor-driven workhead with variable speeds of the taper size specified (see 6.2.1).

c. Other equipment as specified by the procuring activity.

3.9.4 Type IV optional equipment.

a. One internal grinding attachment complete with 1/4-inch collet.

b. One large cutter radius grinding attachment.

c. One 4-inch universal 3-jaw chuck for the universal workhead.

d. Additional index plates for spiral lead indexing unit as specified (see 6.2.1).

e. Other equipment as specified by the procuring activity.

3.10 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All instruction and identification plates shall be of corrosion-resistant material.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval

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Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Optical comparator. When specified (see 6.2.1), an optical comparator of the specified size shall be provided with the machine. The optical comparator shall allow the operator to check tool diameters, lengths, angles, and radii without removing the tool from the tool-holding device. The comparator shall be designed to be used at either the left- or right-hand grinding wheel on dual wheel machines and on swiveling wheelheads at any angle on the column. Means shall be provided to tilt the comparator upwards when not in use. Unless otherwise specified (see 6.2.1), the comparator shall be provided with a cross line screen with protractor ring, tool makers chart, vernier, a 20X lens, a positioning device giving longitudinal and cross adjustment, a power supply, and brackets for mounting the comparator on the machine. The comparator shall be mounted to allow simultaneous viewing of the workpiece and grinding wheel.

3.13 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments no larger than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.14 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.15 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that

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prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under normal operating conditions.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6, all tests in 4.7 and the inspection in 4.8. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

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4.6 Examination. The grinder and equipment shall be examined for design, construction, material, components, electrical equipment, and workmanship to determine compliance with all requirements of this specification as given in 3.1 through 3.15, including compliance with all referenced standards.

4.7 Tests. Unless otherwise specified (see 6.2.1), each machine shall be tested as specified in 4.7.1 through 4.7.4. Failure of any machine to pass any test shall be cause for rejection. All instruments, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the procuring activity in lieu of or in addition to the following tests (see 6.2.1).

4.7.1 Operational test. Each machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms and other accessories, as applicable, shall be provided during the trial period.

4.7.2 Performance test. The performance of each grinder shall be tested as follows for the type of machine specified.

4.7.2.1 Type I and II. Grind or regrind a 4 by 1-1/2 inch, high speed steel, left- or right-hand, arbor mounted type, helical, plain milling cutter having a minimum of 10 cutting teeth. Finished cutter physical dimensions shall conform to the tolerances given in 3.6.

4.7.2.2 Type III and IV machines. Grind or regrind a 1 by 1-5/8 inch, multiple fluted, high speed steel, left- or right-hand, shank mounted type, helical end mill. Shank tapers shall conform to ANSI B5.10. Finished cutter physical dimensions shall conform to 3.6.

4.7.3 Alignment tests. Each machine and its components shall be tested for conformance with the accuracies specified in table III or IV, as applicable. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.12.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951.

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When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use.

6.1.1 Types I and II. The tool and cutter grinders covered by this specification are intended for resharpening conventional types of cutters within the range specified, including all types of milling cutters, reamers, counterbores, hobs, taps, form tools, thread chasers, or other metal-cutting tools. With the proper attachments, they shall be used for internal and external cylindrical grinding and surface grinding.

6.1.2 Types III and IV. The tool, cutter, gashing, flute and radius grinders covered by this specification are intended for resharpening conventional-type end milling cutters having cylindrical or tapered sides, straight or helical flutes, and ball, radius, or square nose. With proper attachments, they shall be used for grinding and sizing shell or ball end mills, face-milling cutters and form-milling cutters that can be mounted on arbors within the capacity of the machine. Type IV machines shall also be used to make new cutters from solid stock.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), stated required system (see 3.2.2).

e. If measuring and indicating devices are required, and any special requirements (see 3.2.2.1).

f. Digital readout system, if required, and any special requirements (see 3.2.2.2).

g. Controls, if different (see 3.2.5).

h. Additional safety and health requirements, if required (see 3.2.6).

i. Painting, if different (see 3.3.5).

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j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.12).

k. Worktable arrangement, if different (see 3.4.2.1, 3.4.2.2., 3.4.2.3, and 3.4.2.4).

l. Automatic cycle on type II machines, if required (see 3.4.2.2).

m. Wheelhead, if different (see 3.4.3.1, 3.4.3.2, 3.4.3.3, and 3.4.3.4).

n. Wheelhead speed on type I machines, if different (see 3.4.3.1).

o. Horizontal grinding wheelhead on type II machines, if required (see 3.4.3.2).

p. Workhead, if different (see 3.4.4.1, 3.4.4.2, and 3.4.4.3).

q. Tailstock, if required; specify taper or internal configuration (see 3.4.6).

r. Coolant system, if required; specify flood or mist type (see 3.4.7).

s. Hydraulic system, if required (see 3.4.8).

t. Electric system and input power, if different (see 3.4.9).

u. Motor enclosure, if different (see 3.4.9.1).

v. Dust collector, if required (see 3.4.10).

w. Size and capacity, if different (see 3.5).

x. Additional performance requirements, if required (see 3.6.1).

y. Standard equipment, if different (see 3.8.1, 3.8.2, 3.8.3, and 3.8.4).

z. Size, number, type, dimensional characteristics, and material composition of grinding wheels (see 3.8.1 (f), 3.8.2 (f), 3.8.3 (d), and 3.8.4 (g)).

aa. Reducing collets, as required (see 3.8.1 (h) and 3.8.2 (h)).

bb. Ball bearing work holding spindle bore required (see 3.8.3 (f)).

cc. Work holding taper size, as required (see 3.8.4 (f)).

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dd. Specify and fully describe optional equipment, if required (see 3.9.1, 3.9.2, 3.9.3, and 3.9.4).

ee. Chuck size, spindle taper, and motor horsepower required for cylindrical grinding attachment for type I machines (see 3.9.1 (a)).

ff. Number of quill sizes and number of grinding wheels, their shank size, type, dimensional characteristics, and material composition required for internal grinding attachment for type I machines (see 3.9.1 (b)).

gg. Number and size of cutter bushings required for gear cutter sharpening attachment for type I machines (see 3.9.1 (d)).

hh. Specify radius and maximum cutter capacity required (see 3.9.1 (e)).

ii. Radius grinding fixture taper configuration, as required (see 3.9.2 (a)).

jj. Workhead taper size, as required (see 3.9.3(b)).

kk. Specify additional spiral lead indexing unit, as required (see 3.9.4 (d)).

ll. Specify applicable repair parts, if required (see 3.10).

mm. Lubrication chart or plate, if different (see 3.11.1).

nn. Nameplate, if different (see 3.11.2).

oo. Optical comparator, if required; specify size and special requirements (see 3.12).

pp. Hour meter, if required (see 3.13).

qq. Inspection conditions, if different (see 4.3).

rr. First article inspection, if different (see 4.4).

ss. Quality conformance inspection, if different (see 4.5).

tt. Test requirements, if different (see 4.7).

uu. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

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6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements Lists, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. Machine classification changes from Federal Specification 00-G-692C are as follows:

00-G-692C		MIL-G-80268	
Type 0,	size 615	Delete	
	size 815	Delete	
Type I, style A,	size 815	Type I, style A,	size 1
	size 1022		size 2
	size 1227		size 3
Type I, style B,	size 815	Type II,	size 1
Type II,	size 815	Type I, style B	size 1
	size 1022		size 2
	size 1227		size 3
Type III,	size 610	Delete	
New		Type III,	size 1
Type IV,	size 84	Delete	
	size 126	Type IV,	size 1

6.5 Subject term (key word) listing.

Abrasive wheel, grinding
Base, floor
Controls, manual, power
Cutting tools, metal
Grinder, cutter, flute, radius, tool
Grinding machine, tool
Machine tool, grinding
Spindle, horizontal, vertical
Wheel, abrasive, grinding

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristics	Type 1					
	Style A			Style B		
Capacity:	Size 1	Size 2	Size 3	Size 1	Size 2	Size 3
Swing over table, inches	8	10	12	8	10	12
Distance between centers, inches						
- of tailstock	15	22	27	15	22	27
- of workhead and tailstock	12	17-1/2	24	11	20	24
- of headstock and tailstock	---	---	---	11	20	24
Distance between centerline of wheel spindle to table top, minimum to maximum, inches	3 to 9-1/2	3-1/4 to 10-1/2	5-3/8 to 13-1/2	3-3/4 to 10-3/4	3-1/4 to 10-1/2	5-3/8 to 13-1/2
Sliding table arrangment:						
Work surface, inches	19 x 4	36 x 5	44 x 6	24 x 4	30 x 5	44 x 6
Longitudinal travel, inches	10	14-1/2	24	10	14-1/2	24
Power or manual table movement	< Manual >			< Manual or Power >		
Power table movement						
Number of feeds	---	---	---	6	7	Infinite
Feedrate, IPM	---	---	---	7 to 100	28 to 90	2 to 200
Cross travel, inches	4-1/2	7-3/4	8	7	8	8
Horizontal swivel, each side of zero, degrees	< 90 >			< 90 >		
Wheelhead or grinding unit:						
Number of spindle speeds	2	2	4	3	2	4
Wheel speed range, RPM	< 3,800 to 5,500 >					
Vertical adjustment of wheelhead, inches	6	7	10-1/2	8	7-1/4	10-1/2
Wheelhead horizontal swivel, degrees, each side of zero	< 110 >			110	106	110
Wheelhead vertical tilt, degrees, each side of zero	< 15 >			< 15 >		
Motor horsepower	1/2	3/4	1	< 1 >		
Workhead or headstock:						
Manual or power	< Manual >			< Manual or Power >		
Number of speeds	---	---	---	4	2	2
Speed range, RPM	---	---	---	128 to 540	128 to 400	265 to 370
Horizontal tilt, degrees, each side of zero	< 90 >			< 90 >		
Motor horsepower	---	---	---	1/6	1/4	1/4

NOTE 1: Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2.1, nor shall they be greater than the next larger size shown in table I above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

NOTE 2: Dashes (---) indicate feature not available on machine.

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TABLE II. Sizes and capacities for type II machines.

Characteristic	Type II, Styles A and B Size 1
Capacity:	
Swing over table, inches	8
Maximum distance between centers, inches	15
Maximum distance between grinding wheel and base table, inches	16-3/4
Maximum cutter diameter, inches	16
Worktable:	
Cross carriage travel, inches	14
Longitudinal table travel, inches	12-1/2
Angular adjustment of carriage, degrees	180
Angular adjustment of subtable in vertical plane, degrees	45
Vertical adjustment of base table, inches	12-1/2
Center table work surface, inches	3-1/2 x 24
Base table work surface, inches	16 x 16
Work holder angular adjustment, inches	180
Wheelhead:	
Spindle speed range, RPM	3,200 to 4,800
Vertical adjustment of grinding wheel, inches	1-1/2
Angular adjustment of spindle head, degrees	
- left	90
- right	70
Maximum horizontal travel of grinding unit, inches	7
Motor horsepower	1

NOTE: Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2.1, nor shall they be greater than the next larger size shown in table II above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE III. Sizes and capacities for types III and IV machines.

Characteristic	Type III	Type IV
Capacity:		
Maximum swing (diameter), inches	12-3/4	12
Maximum tool length, inches	7	6
Maximum radius, inches		
-convex	5-1/2	5
-concave	5-1/2	5
Minimum lead generated by spiral lead mechanism, inches	---	1.81
Workhead:		
Movement normal to cutter axis, total, inches	8-5/8	5-1/4
Movement parallel to cutter axis, total, inches	10-1/8	5-3/4
Turntable base swivel, degrees	210	235
Workhead tilt, degrees	±40	±40
Workhead vertical adjustment, inches	3	---
Wheelhead:		
Longitudinal slide adjustment, inches	7	6
Cross slide adjustment, inches	12-1/2	7-1/2
Vertical slide adjustment, inches	---	9-1/2
Grinding wheel diameter, maximum, inches	6	4
Spindle speeds:		
Number of wheelhead speeds	2	3
Wheelhead, RPM	3,800-5,800	4,000-8,000
Workhead, RPM	---	436
Motors:		
Wheelhead drive, HP	1	1/2
Workhead drive, HP	---	1/4

NOTE 1: Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2.1, nor shall they be greater than the next larger size shown in table III above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

NOTE 2: Dashes (---) indicate a feature not available on the machine.

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TABLE IV. Alignment tests for types I and II.

Test to be applied (as applicable to type of machine)	Maximum, allowable deviation
Worktable:	
Table parallel with wheel spindle	0.002 inch per ft.
Transverse movement of table parallel with table surface	0.0008 inch per ft.
Rise and fall of table in its longitudinal travel	0.0008 inch per ft.
Universal head and tailstock:	
Taper of workhead spindle runs true, 4 inches from nose	0.0003 inch TIR
Tailstock spindle parallel with table	0.0008 inch per ft.
Tailstock spindle parallel with center T-slot of table	0.0008 inch per ft.
Tailstock axis in line with workhead axis (workhead previously to be set parallel with table and T-slot by means of mandrel):	
- in vertical plane	0.0004 inch TIR
- in horizontal plane	0.0004 inch TIR
Wheel spindle:	
Wheel spindle for axial slip and runout	0.0003 inch TIR
Vertical adjustment of wheelhead square with worktable:	
- in plane through long axis of table (for the machines with vertically adjustable wheelhead)	0.0036 inch per ft.
- in plane perpendicular to worktable plane (for machines with vertically adjustable wheelhead)	0.0036 inch per ft.
Wheelhead swivel parallel with table surface	0.002 TIR

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TABLE V. Alignment tests for types III and IV.

Test to be applied	Maximum, allowable deviation
Swivel worktable assembly:	
Parallelism of the top of the worktable with its movement toward and away from the swivel center - in full length of travel	0.001 inch TIR
Parallelism of the top of the worktable with its cross (transverse) movement - in 1-1/4-inch travel (5/8 inch each side of swivel center)	0.001 inch TIR
Parallelism of top of worktable in its total swivel movement and indicating at 6-inch radius from swivel center	0.002 inch TIR
Squareness of the vertical movement of the workhead with the top of the worktable in both directions (full length of travel) (type III machines)	0.002 inch TIR
Wheel spindle:	
Squareness of the vertical movement of the wheel spindle with the top of the worktable in both directions (full length of travel) (type IV machines)	0.002 inch TIR
Parallelism of the cross (transverse) movement of the wheel spindle with top of the worktable, in 4 inches	0.001 inch TIR
Parallelism of the horizontal (axial) movement of the wheel spindle with the top of the worktable, in 4 inches	0.001 inch TIR
Concentricity of the wheel spindle:	
- Straight diameter	0.002 inch TIR
- Face	0.001 inch TIR

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Navy - SH
Air Force - 99

Review activities:
Army - ME
Air Force - 84
DLA - GS

User activities:
Army - SM, MU, AR
Navy - MC
Air Force- None

Preparing activity:
DLA - IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80268	Grinding Machines, Tool and Cutter; and Grinding and Gashing Machines, Tool, Cutter, Flute and Radius

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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